

SAUDA, Norway

WMO: 014240

Lat: 59.649N

Lon: 6.363E

Elev: 6

StdP: 101.25

Time Zone: 1.00 (EUC)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.8	-9.9	-16.5	0.9	-9.0	-14.8	1.0	-7.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.6	26.0	16.0	23.5	15.5	21.5	14.5	17.7	23.2	16.7	21.4	15.7	19.8	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.7	11.1	19.2	14.7	10.4	18.3	13.8	9.8	17.4	49.8	23.0	46.8	21.7	44.0	19.9	21.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
N/A	N/A	N/A	-11.8	29.9	2.5	3.1	-13.6	32.1	-15.0	33.9	-16.4	35.6	-18.2	37.9		
			-12.4	19.3	2.3	1.4	-14.0	20.3	-15.3	21.1	-16.6	21.9	-18.2	23.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.5	-0.5	-0.2	2.6	6.9	11.0	13.7	16.4	15.3	12.0	7.5	4.0	0.3
	DBStd	6.81	4.01	3.83	2.99	3.13	3.98	2.77	3.09	2.28	2.50	3.07	3.33	4.40
	HDD10.0	1558	326	284	229	103	34	2	0	0	9	88	181	300
	HDD18.3	3995	584	518	487	343	230	142	78	97	189	337	430	558
	CDD10.0	634	0	0	1	10	66	114	198	166	70	9	1	0
	CDD18.3	28	0	0	0	0	3	3	18	4	0	0	0	0
	CDH23.3	225	0	0	0	0	33	36	134	22	0	0	0	0
	CDH26.7	45	0	0	0	0	7	2	34	2	0	0	0	0
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	2412	300	244	213	130	105	110	129	176	201	248	260	287
	PrecMax	3252	612	530	583	265	257	246	214	325	374	532	493	618
	PrecMin	1538	20	7	9	43	5	16	23	44	43	43	27	15
	PrecStd	436	151	147	122	56	63	67	46	68	112	127	133	169
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.6	9.0	11.9	19.2	27.3	26.7	30.0	25.9	20.5	18.0	12.8	10.2
		MCWB	4.7	5.1	7.2	10.0	14.5	15.4	18.2	16.3	13.6	11.8	8.7	6.7
	2%	DB	6.8	7.3	9.2	17.3	23.3	24.2	27.0	23.0	19.0	14.2	11.3	8.6
		MCWB	4.4	4.1	5.0	9.3	13.7	14.7	17.5	15.6	12.9	11.4	7.0	5.8
	5%	DB	5.6	5.9	8.0	14.9	20.9	21.2	24.1	21.2	17.7	12.9	10.1	7.2
		MCWB	3.0	3.5	4.6	8.2	12.4	13.6	16.8	14.8	12.2	9.8	7.4	4.6
	10%	DB	4.4	4.6	6.9	12.4	18.3	19.1	21.7	19.5	16.4	11.7	8.8	6.0
		MCWB	2.0	2.8	4.2	7.0	11.4	12.6	15.6	14.4	12.2	8.8	7.0	3.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.0	6.4	7.4	11.1	16.0	16.9	19.5	18.3	15.5	13.7	9.9	8.1
		MCDB	7.0	8.0	10.4	16.9	24.6	23.5	26.0	23.1	17.3	15.1	11.6	9.3
	2%	WB	4.7	4.8	6.3	9.7	14.4	15.4	18.3	16.9	14.3	12.1	9.0	6.3
		MCDB	6.5	6.5	8.0	15.7	22.1	21.6	24.5	20.5	16.8	14.0	10.2	7.9
	5%	WB	3.5	3.8	5.3	8.7	12.9	14.2	17.4	16.0	13.6	10.7	7.8	5.0
		MCDB	5.1	5.4	7.2	13.8	19.7	19.6	22.7	19.2	16.0	12.4	9.1	6.8
	10%	WB	2.5	3.0	4.5	7.6	11.7	13.3	16.4	15.2	12.8	9.4	6.7	3.9
		MCDB	3.8	4.3	6.5	11.6	17.5	18.1	21.2	18.2	15.1	10.9	8.4	5.5
Mean Daily Temperature Range	5% DB	MDBR	4.8	5.4	6.1	8.1	9.0	8.4	7.6	7.1	6.3	5.5	4.7	4.5
		MCDBR	6.0	5.8	7.0	12.2	13.6	12.8	12.2	10.0	8.6	6.6	5.9	5.8
		MCWBR	4.1	3.7	4.4	6.3	6.0	5.8	4.9	4.4	4.4	4.1	4.1	4.3
	5% WB	MCDBR	5.3	5.3	6.2	10.6	12.5	10.4	10.7	7.8	6.9	5.4	5.4	5.7
		MCWBR	4.0	3.9	4.5	5.6	5.8	5.4	4.7	4.2	4.2	4.1	4.3	4.5
Clear-Sky Solar Irradiance	taub	0.275	0.309	0.343	0.383	0.381	0.373	0.390	0.384	0.358	0.334	0.290	0.255	
	taud	2.104	2.175	2.160	2.131	2.262	2.384	2.366	2.375	2.440	2.385	2.198	2.044	
	Ebn at Noon	491	682	774	808	845	859	833	807	770	662	476	362	
	Edh at Noon	46	75	103	127	120	109	108	100	80	62	42	33	
All-Sky Solar Radiation	RadAvg	0.31	0.92	2.19	4.00	4.74	5.06	4.56	3.56	2.28	1.11	0.41	0.20	
	RadStd	0.07	0.21	0.45	0.71	0.79	0.62	0.59	0.31	0.23	0.18	0.08	0.03	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page